

Aayush Kumar

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Education

2026 –  **Ph.D. in Computer Science**
Massachusetts Institute of Technology, Boston, USA.

2020 – 2024  **B.Tech. in Computer Science and Engineering**
IIT Kanpur, India

Employment History

2024 – 2026  **Research Fellow**, PROSE team, Microsoft, India.

2023  **Quantitative Research Intern**, Quadeye Securities LLP, India.

Publications

Conference Papers

- 1 **A. Kumar**, Y. Bajpai, S. Gulwani, G. Soares, and E. Murphy-Hill, “Why AI Agents Still Need You: Findings from Developer-Agent Collaborations in the Wild,” in *2025 40th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2025, pp. 432–444.  DOI: 10.1109/ASE63991.2025.00043.
- 2 J. T. Liang, **A. Kumar**, Y. Bajpai, S. Gulwani, V. Le, C. Parnin, A. Radhakrishna, A. Tiwari, E. Murphy-Hill, and G. Soares, “TableTalk: Scaffolding Spreadsheet Development with a Language Agent,” *ACM Trans. Comput.-Hum. Interact.*, vol. 32, no. 6, Dec. 2025, ISSN: 1073-0516.  DOI: 10.1145/3765286.
- 3 N. Mehrotra, **A. Kumar**, S. Gulwani, A. Radhakrishna, and A. Tiwari, “TEN: Table explicitization, neurosymbolically,” in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*, Y. Li, G. Rehm, and M. Tu, Eds., San Diego, California, USA: Association for Computational Linguistics, Jul. 2026, pp. 2050–2086, ISBN: 979-8-89176-394-4.  DOI: 10.18653/v1/2026.acl-industry.138.
- 4 H. Goel, **A. Kumar**, and S. S. Ragavan, “End-User Programming is WEIRD: How, Why and What to Do About It,” in *2023 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, 2023, pp. 41–50.  DOI: 10.1109/VL-HCC57772.2023.00013.
- 5 M. Amoozadeh, D. Daniels, D. Nam, **A. Kumar**, S. Chen, M. Hilton, S. Srinivasa Ragavan, and M. A. Alipour, “Trust in Generative AI among Students: An exploratory study,” in *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1*, ser. SIGCSE 2024, Portland, OR, USA: Association for Computing Machinery, 2024, pp. 67–73, ISBN: 9798400704239.  DOI: 10.1145/3626252.3630842.

Pre-prints

- 1 **A. Kumar**, D. Prol, A. Alipour, and S. S. Ragavan, *To Google or To ChatGPT? A Comparison of CS2 Students’ Information Gathering Approaches and Outcomes*, 2025.  URL: <https://arxiv.org/abs/2501.11935>.
- 2 **A. Kumar** and S. Mhatre, *UnWEIRDing LLM Entity Recommendations*, 2025.  URL: <https://arxiv.org/abs/2511.18403>.

Conference Posters

- 1 M. Amoozadeh, D. Daniels, S. Chen, D. Nam, **A. Kumar**, M. Hilton, M. A. Alipour, and S. S. Ragavan, "Towards Characterizing Trust in Generative Artificial Intelligence among Students," in *Proceedings of the 2023 ACM Conference on International Computing Education Research - Volume 2*, ser. ICER '23, Chicago, IL, USA: Association for Computing Machinery, 2023, pp. 3–4, ISBN: 9781450399753. [DOI](#): 10.1145/3568812.3603469.
- 2 H. Goel, **A. Kumar**, and S. S. Ragavan, "Poster: End-User Programming is WEIRD," in *2023 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, 2023, pp. 274–275. [DOI](#): 10.1109/VL-HCC57772.2023.00051.

Awards and Achievements

- 2025  **Outstanding Reviewer**, iTiCSE 2025.
- 2024  **Research Proficiency Medal**, IIT Kanpur, India.
Awarded for best undergraduate research project by a graduating student in the Computer Science and Engineering department.
- 2022  **Academic Excellence Award**, IIT Kanpur, India.
- 2021  **Dr. Prateek Mishra Memorial Scholarship Awardee**, IIT Kanpur, India.
Awarded for achieving the highest GPA in my department during freshman year.
 **Academic Excellence Award**, IIT Kanpur, India.
- 2020  **All India Rank 976**, Joint Entrance Examination - Advanced, India.
 **All India Rank 4128**, Joint Entrance Examination - Mains, India.

Reviewing Experience

- 2026  **Reviewer**, VL/HCC 2026
 **PC Member**, BiAlign Workshop on Human-AI Interaction Alignment, CHI 2026
 **PC Member**, iTiCSE 2026
- 2025  **External Reviewer**, TEI 2026
 **Sub-Reviewer**, ASE 2025
 **Reviewer**, iTiCSE 2025
- 2023  **Sub-Reviewer**, ICSE 2024

Invited Talks

- 2026  **EuSpRIG 2026**
Keynote: Towards Collaboration-First AI for End-User Programming

Teaching Experience

- 2023 - 2024  **Tutor**, ESC111: **Fundamentals of Computing**, CSE Department, IIT Kanpur, India.
Taught a CS1 course to a class of 25-30 students once a week during both the Fall and Spring semester in my senior year.
- 2022  **Volunteer**, Prayas (Student-run Community Service Organisation), IIT Kanpur, India.
Taught mathematics to underprivileged grade 12 students

Selected Projects

- 1 **Intersectionality in Oversampling on Social Attributes**, Semester Project, ML4ED Lab, EPFL, 2023, Built on the paper by Cock et al. to analyze the impact of intersectionality on different oversampling strategies for a behavioural predictive model in education.
- 2 **SoulFull: An AI-Powered Companion App for Binge Eating**, Course Project, EPFL, 2023, Designed a high-fidelity prototype for an app that supports people struggling with binge eating via journalling and AI-powered insights. 🔗 URL: <https://www.figma.com/file/0LqV3BIDvyHij9iheE4esL/>.
- 3 **SUTRA Jupyter Implementation**, Summer Project, C3iHub, IIT Kanpur, 2022, Worked on designing a web application for the SUTRA COVID predictive model with Dr. Manindra Agarwal. Helped translate the SUTRA Mathematical Model from Excel to Jupyter code with user inputs and parameters. 🔗 URL: <https://github.com/Aeyush10/SUTRA>.
- 4 **DigiCampus**, Course Project, IIT Kanpur, 2022, Worked with a team to create a web application to simplify entry-exit logistics for students and faculty across college campus. 🔗 URL: https://github.com/Aeyush10/CS253_Project.